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SUPERSEDING
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FEDERAL SPECIFICATION

FIXTURE, LIGHTING (FLUORESCENT, ALTERNATING-CURRENT, RECESSED AND SURFACE CEILING)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers recessed and surface ceiling mounted fluorescent lamp lighting fixtures (luminaires) for alternating current service, as specified (see 3.2.4). Unless otherwise specified (see 6.1), lamps shall not be included with the fixtures.

1.2 Classification. Lighting fixtures (luminaires) furnished under this specification shall be of the direct lighting style in accordance with requirements of 3.5.1, and shall be of the following classes, types, sizes, mounting, and lamp arrangements, as specified (see Table I and 6.1):

Classes:

Class I - Recessed, Air-handling - recessed fixture (fixture mounted within a ceiling) designed for cooled, heated or ventilated air delivery, air removal, and heat exchange (lamp and ballast cooling). Suitable for lighting offices, classrooms, laboratories, and other areas in which prolonged or critical visual tasks are to be performed.

Class II - Recessed, Static - recessed fixture not having air-handling capability. Suitable for lighting offices, classrooms, laboratories and other such areas in which prolonged or critical visual tasks are to be performed.

Class III - Recessed, Restricted use - non air-handling recessed fixture having dropped shape white opal diffuser. Suitable for lighting hallways, lobbies, lounges, restrooms and other such spaces in which prolonged or critical visual tasks are not normally performed.

Class IV - Surface Ceiling - non air-handling fixture designed for mounting against or within one and one half (1 1/2) inches of the surface of ceilings. Suitable for lighting offices, classrooms, laboratories and other such areas in which prolonged or critical visual tasks are to be performed.

Class V - Surface Ceiling, Restricted use - non air-handling surface ceiling fixtures having dropped shape white opal diffuser. Suitable for lighting hallways, lobbies, lounges, restrooms and other such spaces in which prolonged or critical visual tasks are not normally performed.

Types (Applicable to Class I, II and III fixtures only):

Type F - Recessed luminaire having horizontal flanges parallel to the lamp direction, designed to conceal the edges of the ceiling opening, above which the luminaire is supported, by concealed mechanical suspension.

Type M - Recessed luminaire having vertical turned-up edges parallel to the lamp direction and intended to "snap-in" or otherwise align the luminaire with a conceiled T-bar suspension system, the center opening of the Tees being located on modular or other symmetrical dimensional lines.

Type G - Recessed luminaire designed to rest on or "lay-in" the exposed inverted T of a suspension system (customarily described as a "grid" ceiling system) with the webs of the Tees being located on modular or other symmetrical dimensional lines.

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Type S - Recessed luminaire designed for mechanical suspension from exposed splines and dependent on splines parallel to the lamp direction for concealment of the edges of the luminaire.

Type HS - A type H recessed luminaire having edges parallel to the lamp direction and dependent on splines of the ceiling suspension system for concealment of the edges of the luminaire. A Type H luminaire is a recessed luminaire having end brackets hooks or other attachments and designed to be supported at the ends by "hooking-on" to some member of the ceiling suspension system.

Type HF - A Type H recessed luminaire havind edges parallel to the lamp direction and designed to conceal the edges of the ceiling opening in which luminaire is recessed.

NOTE: The Types cited above are National Electrical Manufacturers Association (NEMA) type designations and, except as may be otherwise specified herein, shall be in accordance with NEMA Pub. No. LE 1 - 1969. Refer to NEMA Pub. No. LE 1 - 1969 for detailed description of the above Types.

Sizes

Size 1 - 1 foot x 4 feet (in Lamp Arrangements A only, except as noted below under description of Lamp Arrangement B).

Size 2 - 2 feet x 4 feet (in Lamp Arrangements A, B, or C only).

Size 4 - 4 feet x 4 feet (in Lamp Arrangements C. D, or E only).

Mountings:

Mounting 1 - Individually mounted

Mounting 2 - Continuous Row mounted

Lamp Arrangements:

Arrangement A - Two 40-watt, medium bi-pin base, 48 inch, T-12 rapid start fluorescent lamps. (Applicable to size 1 and 2 fixtures only).

Arrangement B - Three 40-watt, medium bi-pin base, 48 in, T-12 rapid start fluorescent lamps. (Applicable to class III and V, size 1 fixtures and all size 2 fixtures).

Arrangement C - Four 40-watt, medium bi-pin base, 48 inch, T-12 rapid-start fluorescent lamps. (Applicable to size 2 and 4 fixtures only).

Arrangement D - Six 40-watt, bi-pin base, 48 inch, T-12 rapid-start flurorescent lamps. (Applicable to size 4 fixtures only).

Arrangement E - Eight 40-watt, medium bi-pin base, 48-inch, T-12 rapid-start fluorescent lamps. (Applicable to size 4 fixtures only).

NOTE: The above mentioned lamps shall be construed as being in conformance with ANSI Standard C78.700.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

W-B-30 - Ballast, Fluorescent Lamp
PPP-B-585 - Boxes, Wood, Wirebound
PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock-corner
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple Wall

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Federal Standards:

Fed. Std. No. 123	-Marking for Domestic Shipment (Civil Agencies).
Fed. Test Method Std. No. 151	-Metals: Test Methods.
Fed. Test Method Std. No. 406	-Plastics; Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking For Shipment and Storage.
MIL-STD-461 - Electromagnetic Interference Characteristics, Requirements For Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-463 - Definitions and Systems of Units, Electromagnetic Interference Technology.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Electrical Manufacturers Association (NEMA) Publications:

Pub. No. LE 1-1969 - Fluorescent Luminaires.

(Application for copies should be addressed to National Electrical Manufacturers Association, 155 East 44th Street, New York, N.Y. 10017.)

Underwriters' Laboratories (UL), Inc. Standards:

No. 57 - Electric Lighting Fixtures.
No. 542 - Lampholders, Starters, and Starter Holders for Fluorescent Lamps.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Illinois 60611; 1285 Walt Whitman Road, Melville, L.I., New York 11746; or 1655 Scott Blvd., Santa Clara, California 95050.)

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Illuminating Engineering Society (IES) Publications:

IES Lighting Handbook, Fifth Edition 1972.

RQQ Report No. 2 (1972) - Outline of a Standard Procedure for Computing Visual Comfort Ratings for Interior Lighting (IES Reprint LM-42).

Journal of IES, Volume 1, No. 2, January, 1972, Determination of Average Luminance of Luminaires (IES Reprint LM-37).

(Application for copies should be addressed to the Illuminating Engineering Society, 345 East 47th Street, New York, New York 10017.)

Air Diffusion Council Test Codes:

Air Diffusion Troffer Heat Removal Test Code TC-66R1

Equipment Test Code 1062R3

(Application for copies should be addressed to the Air Diffusion Council, 435 North Michigan Ave., Chicago, Illinois 60611.)

American National Standards Institute (ANSI), Inc. Standards:

No. C78.700 - 40-Watt T-12 Rapid-Start Fluorescent Lamp.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, N.Y. 10018.)

American Society for Testing and Materials (ASTM) Standard:

No. D-788 - Standard Specification For Methacrylate Mouldings and Extrusion Compounds.

No. D-702 - Tentative Specification for cast Methacrylate Plastic Sheets, Rods Tubes and Shapes.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Inc., Agent, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libaries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS.

3.1 Fire, safety and casualty hazards.

3.1.1 Each contractor shall submit to the contracting agency proof that each fixture he proposes to supply under this specification conforms to the requirements of UL 57 - Electric Lighting Fixtures. Listing by the UL may be accepted as evidence that the product conforms to this requirement. Compliance with the above preliminary requirements in regard to fire, safety and casualty hazards does not absolve the contractor from complete compliance with the other requirements of this specification in order to secure acceptance of his material or equipment.

3.2 Design, general.

3.2.1 Accessibility of parts. With the exception of Size 1, three lamp fixtures, lamps shall be removable and replaceable without the removal of other lamps. All fixtures shall be designed such that lamp installation/removal can be accomplished without tools. The ballast(s) and wiring on all fixtures shall be easily accessible without the use of special purpose tools. Also, ballast(s) and lamp holders must be installed in a manner that will allow replacement without removal of the fixture from the ceiling.

3.2.2 Lamp spacing. For lamp arrangement A, the lamps shall be equidistant from the center of the fixture; for lamp arrangement B, the middle lamp shall be on the centerline of the fixture and the two outside lamps shall be equidistant from the center of the fixture; for lamp arrangement C, the two outside lamps shall be equidistant from the center of the fixture. Arrangements D and E shall have uniform arrangement similar to arrangement C. With exception of Size 1, three lamp fixtures, each lamp shall be sufficiently spaced to permit easy removal and replacement by hand without disturbing the other lamps.

3.2.3 Lampholders. The lampholders shall be medium bi-pin type in accordance with the applicable requirements of UL 542. Lampholders shall hold the lamps securely without the use of auxiliary devices. Lampholders shall be securely fastened to the mounting surface with necessary provisions to prevent the lampholders from turning. The lampholders and mountings shall be sufficiently rigid to prevent spreading upon installation of the lamp. The longitudinal distance between the inner surfaces of a pair of lampholders shall be sufficient, but not greater than necessary to insure proper fit and contact of the ANSI C78.700 lamps specified herein. All electrical contact surfaces of lampholders shall insure consistent starting and operation of the lamp.

3.2.4 Electrical requirements. Fixtures shall be designed to operate on one of the following voltage ranges and alternating current frequencies as specified (see 3.7 and 6.1): 110-125 volts, 60 Hertz; 254-290 volts, 60 Hertz; or 210-236 volts, 50 Hertz.

3.3 Design, detailed.

3.3.1 Recessed fixtures. Recessed fixtures shall be designed for mounting within a ceiling in such manner that bottom opening of luminaire housing is flush with lower surface of ceiling, recessed into ceiling no more than .070 inch or extending below ceiling no more than .150 inch. Recessed fixtures covered herein shall be of the following designs, designated as Class I, II or III.

3.3.1.2 Class I. Class I fixtures shall be designed to illuminate office, classroom, laboratory, and other such spaces. Additionally, when used in conjunction with appropriate air handling equipment (air diffusers, dampers and pattern controls) Class I fixtures shall perform in accordance with the air handling function(s) and mode(s) of operation cited herein. The particular type of air handling function shall be in accordance with UL 57 Classifications A,B,C,D,E Or F for air handling fixtures, and the desired modes of operation shall be as cited below. Both the UL 57 air handling function (A,B,C,D,E, or F) and mode(s) of operation shall be indicated by specification user (see 6.1). The modes of operation shall be as follows:

1. Static - not handling either supply or return air.

2. Air supply, single side - supply air to occupied space through single side slot.

3. Air supply, saddle or twin - supply air to occupied space through two side slots.

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4. Air return through lamp cavity (for heat exchange) - return air from occupied space through end slots and lamp cavity to plenum or ducts for purpose of heat exchange (lamp and ballast cooling).

5. Air return through side slots - air return from occupied space, through one or both side slots, to plenum or ducts.

6. Air return through lamp cavity (for heat exchange) and side slots - air return from occupied space through end slots and lamp cavity (for heat exchange) and through one or both side slots to plenum or ducts.

7. Air supply, single side, and air return through lamp cavity (for heat exchange) - supply air to occupied space through one side slot and return air through end slots and lamps cavity (for heat exchange) to plenum or ducts.

8. Air supply saddle or twin,, and air return through lamp cavity (for heat exchange) - supply air to occupied space through both side slots and return air through end slots and lamp cavity (for heat exchange) to plenum or ducts.

9. Air supply one side and air return other side - supply air to occupied space through one side slot and return air through second side slot to plenum or ducts.

Refer to part 2 of NEMA Pub. No. LE 1 - 1969 for a more detailed description of modes of operation.

3.3.1.2.1 Air supply, air return and heat exchange operation. For air supply, luminaires shall be designed such that supply air can be piped to side slot(s), from building air ducts and air diffusers, without passing through lamp compartment. For air return the design shall be such that return air passes through side slot(s) and directly to plenum or ducts, without passing through lamp compartment. For heat exchange the design shall be such that return air passes into end slots and through lamp compartment to plenum or ducts, for heat exchange.

3.3.1.2.2 Supply/return and heat exchange (independent operation of). Luminaires shall be designed such that the air supply, air return, and heat exchange operations cited in 3.3.1.2.1 are independent of each other in a manner that will permit any operation to be omitted or included without affecting the others.

3.3.1.2.3 Slot location. Side slots shall be located parallel to each other, adjacent to and along each length dimension edge. End slots shall be located parallel to each other, adjacent to and along each width dimension edge. On size 4 fixtures, where there may be no distinction between length and width dimensions, side slots shall be parallel to longitudinal axis of lamps (when installed), adjacent to and along two edges of fixture; end slots shall be located parallel to each other, adjacent to and along the other two edges of fixture. The edges along which side slots are located shall be construed as fixture sides, and edges along which end slots are located shall be construed as ficture engs. Slot openings shall be either recessed or flush with the bottom of fixture housing. Unless otherwise specified (see 6.1), accessory or integral covers shall be provided with each fixture to close off side slots when not being used for air delivery, or air return.

3.3.1.2.4 Air diffurers (supply and return air). Each supply air diffuser shall be compatible with fixture and ducting system utilized. When specified, supply air diffusers shall be of an insulated design with insulation thickness as required (see 6.1). Air deffusers used in conjunction with ducted return air or heat exchange operations shall be of appropriate construction, compatible with the fixture and ducting system utilized. Types of ducting system(s) to be accomodated shall be specified (see 6.1).

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3.3.1.2.5 Dampers. Dampers for controlling the volume of air supplied and air returned when operating in the supply mode or heat exchange mode shall be provided. Dampers for controlling supply air volume shall be incorporated in the air diffuser or fixture design. Dampers for controlling return air volume, for return through end slots (for heat exchange), shall be incorporated in the fixture design. Dampers shall be capable of controlling volume of air flow through the range from approximately zero air flow to maximum volume air flow for which diffuser and/or luminaire is designed.

3.3.1.2.6 Pattern controls. Pattern controls for controlling the pattern and direction of supply air when operating in any of the modes specified herein shall be incorporated in the air diffuser or fixture design.

NOTE: Air handling equipment, i.e., air diffusers, dumpers, and pattern controls, as discussed in 3.3.1.2.4 thru 3.3.1.2.6 above, that is not an integral part of fixture design shall be furnished separately by manufacturers (or suppliers) of such equipment, unless user of this specification indicates that such equipment is to be furnished by fixture manufacturer (or supplier). (See 6.1).

3.3.1.2.7 Air supply, air return, heat exchange and noise criteria requirements. The following air supply, air return, heat exchange and noise criteria requirements for each type, size, mounting, and lamp arrangement of Class I fixtures shall be indicated by persons using this specification (see 6.1).

(1) Volume of air flow in cubic feet per minute (c.f.m.) for air supply, air return, or heat exchange operation.

(2) Maximum static pressure drop, at volume of air flow (c.f.m.) required in (1) above, for air supply, air return, and heat exchange operation.

(3) Supply air throw and spread for given supply temperature difference and terminal velocity, at the volume of air flow (c.f.m.) required in (1) above.

(4) Heat removed (in Btu/hour) for a given room temperature at volume of air flow (c.f.m.) required in (1) above during heat exchange operation.

(5) Relative light output (RLO) at volume of air flow (c.f.m.) required in (1) above during heat exchange operation.

(6) Noise criteria for volume of air flow (c.f.m.) required in (1) above for specified mode(s) of operation (see 3.3.1.2).

Note: Testing to determine compliance with these requirements shall be in accordance with 4.7.

3.3.1.3 Class II. Class II fixtures shall be designed for illumination of office, classroom, laboratory and other such spaces and shall not have air handling or heat exchange capability.

3.3.1.4 Class III. Class III fixtures shall be designed for illumination of spaces in which prolonged and/or critical visual tasks are not normally performed. ClassIII luminaires shall be designed with a dropped shape white opal diffuser extending below fixture housing. Class III fixtures shall not have air handling or heat exchange capability.

3.3.2 Surface ceiling fixtures (Class IV and V). Surface ceiling fixtures shall be designed for mounting directly against ceiling or for mounting within one and one half (1 1/2) inches of the ceiling with spacers that are integral to the luminaire. When specified (see 6.1), class IV and V fixtures shall be designed for mounting to ceiling composed of combustible low-density sheet or tile as defined by the National Electrical Code, lastest edition.

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3.3.2.1 Class IV. Class IV fixtures shall be designed for illumination of office, classroom, laboratory, and other such spaces and shall not have air handling or heat exchange capability.

3.3.2.2 Class V. Class V fixtures shall be designed for illumination of spaces in which prolonged and/or critical visual tasks are not normally performed. Class V luminaires shall be designed with a dropped shape white opal diffuser extending below fixture housing. Class V fixtures shall not have air handling or heat exchange capability.

3.4 Construction. The construction of fixtures covered herein shall conform to applicable requirements of UL 57.

3.4.1 Grounding device. The housing of each fixture shall be provided with a separate, factory-installed grounding device. The grounding device shall be designed for connection of a separate grounding conductor to the fixture housing, and for no other purpose. The device shall meet all applicable grounding requirements of the National Electrical Code, latest edition.

3.4.2 Wiring. Wiring between fluorescent lampholders and associated operating and starting equipment shall be of sizes not smaller than the sizes of the leads furnished with the required types of ballasts and shall have equal or better insulating and heat resisting characteristics. All wiring within the fixture and from fixture to the splice with building wiring shall conform to the requirements of the National Electrical Code, latest issue, and UL Standard 57. Wiring shall be protected at all points where abrasion is liable to occur and shall be concealed within fixture.

3.4.2.1 Connections. Connections of wires to terminals of lampholers shall be electrically and mechanically secure, with no loose strands protruding. Additionally, the number of wires extending to or from the terminal(s) of a lampholder or other accessory shall not exceed the number for which the terminal(s) or accessory was designed.

3.4.2.2 Joints. Joints is wiring within lighting fixture shall be accomplished in such a manner as to provide conduction and insulation characteristics equivalent to characteristics of wiring being joined.

3.4.2.3 Channels and wireways. Wiring channels and wireways shall be free from projections and rough or sharp edges. All points or edges over which conductors must pass shall be rolled or bushed in accordance with U. 57. Adequate number and sizes of knockouts shall be provided to facilitate connection to building wiring.

3.4.3 Housing or chassis. Fixture housing or chassis shall be constructed of either 20 gauge (0.032 inch, uncoated) steel or 22 gauge (0.026 inch, uncoated) steel with stiffening ribs, bends or embossing, or 0.040 inch minimum thickness (uncoated) aluminum. Additionally, fixture housing hall not have panels, plates or any other type baffle extending more than one inch below bottom closure lens frame, either as an integral part of the housing or as an add on feature.

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3.4.4 Bottom closures and side panels.

3.4.4.1 Class I, II and IV. Class I, II, and IV luminaires shall have flat bottom closures of the lens type, with lens being self supporting and of one piece construction. Bottom closures shall be either flush with or regressed into bottom of fixture housing. The lens of bottom closure shall be of plastic or glass and shall have a uniform overall pattern or prisms or spherical segments on either one or both sides. Side panels, if provided on Class IV luminaires, shall also be of the lens type and constructed of plastic or glass. All plastic lenses shall be of acrylic, meeting ASTM D-788 requirements (see 3.8). All glass side panels, and size 1 and 2 luminaire glass lenses shall be annealed, unless tempered glass is specified. If tempered glass is desired, the specification user shall so indicate (see 6.1). On size 4 luminaires, all glass lenses shall be tempered. The overall cross sectional thickness of size 1 and 2 bottom closure lenses shall be a nominal 0.125 inch (minimum 0.115 inch) if constructed of plastic; and a nominal 0.160 inch (minimum 0.150 inch) if of glass. Size 4 bottom closure lenses shall be a nominal 0.187 inch (minimum 0.182 inch) if constructed of plastic; and a nominal 0.205 inch (minimum 0.195 inch) if of glass. Side panel lens thickness shall be a nominal 0.125 inch (minimum 0.115 inch) if of plastic; and a nominal 0.160 inch (minimum 0.150 inch) if of glass. Bottom closure lens sag - the vertical distance that bottom center of lens sags below lens peripheny, when fixture with bottom closure installed is mounted in a normally functioning position - shall not exceed 1/8 inch for size 1 fixtures; 1/4 inch for size 2 fixtures; and 1/2 inch for size 4 fixtures. Bottom closure and engaging latches shall be constructed in such manner as to preclude accidental (unintentional) release or disassembly of bottom closure, or portions thereof, from fixture housing when in either the latched or open position. The fixture housing and bottom closure shall incorporate a light trap for preclusion of light leaks. Additionally, lens frame(s) shall be constructed in such manner as to permit the replacement of lenses without the use of tools other than screwdrivers or pliers. Also, lens frame shall be sufficiently rigid so as to preclude sag of more than 0.047 below luminaire housing flanges. Bottom closure and side panel lenses may be tinted, provided that photometric and other requirements of this specification are adhered to.

3.4.4.2 Class III and V. Bottom closure construction shall be as specified in 3.4.4.1, except that class III and V bottom closures shall have a dropped shape white opal diffuser fabricated from 0.100 inch flat sheet acrylic plastic. The diffuser shall have a prestressed arch for preclusion of sag, and must meet ASTM D-702 or D-788 requirements (see 3.8).

3.4.5 Mounting. Mounting 1 shall be construed as mounting wherein luminaires are mounted separately, independent of each other. Mounting 2 shall be construed as mounting wherein luminaires are mounted in a continous row, adjacent end to end.

3.4.5.1 Recessed luminaires (class I, II, III). Fixture shall be furnished with all necessary mounting hardware and shall be constructed such that ceiling installation, as applicable for fixture type, size and mounting (see 1.2) can be readily effected. Additionally, fixture construction and mounting hardware shall be such that connection to ceiling support system can be accomplished at no less than four points. For type G luminaires, the mounting shall be such that luminaire is either locked to the inverted "T" at no less than 4 points around luminaire periphery or independently connected to building structural ceiling at no less than 4 points. Mounting design for all fixtures shall be such as to preclude accidental desassembly (falling) of luminaire from ceiling. Specific mounting requirements, if required, shall be indicated in 6.1. Otherwise, fixture construction and mounting hardware, shall be the manufacturers option provided that all applicable requirements of this specification are adhered to.

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3.4.5.2 Surface ceiling luminaires (class IV and V). Mounting design shall be such as to preclude accidental disassembly (falling) of luminaire from ceiling. Size 1 and 2 surface ceiling fixtures shall be constructed such that secure mounting, as applicable for mounting 1 or 2, to ceiling can be effected at no less than (4) points. Size 4 surface ceiling fixtures shall be constructed such that mounting in the above manner can be effected at no less than six (6) points. All necessary hardware to accomplish mounting in the above manner shall be furnished. Mounting 2 fixtures shall be constructed such that continuous wireways can be readily formed when two or more luminaires are joined. Each mounting 2 luminaire shall be provided with a means whereby alignment and electrical continuity with other fixtures can be readily obtained. Each mounting 1 luminaire shall be provided with two end plates, either integral with fixture or of a design that can be secured to fixture at time of installation. One detachable end plate shall be provided with each mounting 2 fixture. Specific mounting requirements, if required, shall be indicated in 6.1. Otherwise, fixture construction and mounting hardware shall be the manufacturers option provided that all applicable requirements of this specification are adhered to.

3.4.6 Dimensions. Overall length and width dimensions shall be as cited below for the appropriate fixture sizes (see 1.2) covered herein. Unless otherwise specified, (see 6.1), length and width tolerance for class I, II, and III (recessed fixtures) shall be + 0, -1/32 inch, and width dimension tolerance for class IV and V luminaires shall be + 1/4 inch. Also, unless otherwise specified (see 6.1), height dimensions for all luminaires covered herein, as measured from lowest portion of bottom closure to top of fixture housing shall be the manufacturer's option provided all applicable specification requirements are adhered to. Overall length and width dimensions for fixtures covered herein shall be:

Size 1 - 1 foot wide by 4 feet long
Size 2 - 2 feet wide by 4 feet long
Size 4 - 4 feet wide by 4 feet long

Note: Length dimensions shall be construed as those luminaire dimensions parallel to longitudional axis of lamp(s). Width dimensions shall be construed as those luminaire dimensions perpindicular to lamp(s) longitudinal axis, lying in a horizontal plane.

3.5 Material. Materials used in construction of fixtures shall be free from all defects and imperfections which might affect serviceability or appearance.

3.6 Parts and assembly. When specified (see 6.1), fixture manufacturer shall furnish all necessary air handling equipment required for operation as indicated in 3.3.1.2. Otherwise, the air handling equipment, required for operation as indicated in 3.3.1.2, shall be furnished separately. Fixture installation hardware and all parts necessary to constitute a complete functional luminaire shall be furnished by the fixture manufacturer. Each fixture shall be assembled and ready for installation and connection to building wiring, except that installation hardware and bottom closure may be shipped disassembled from luminaire.

3.7 Ballast. Ballast(s) shall conform to Federal Specification W-B-30 and shall be designed to operate on one of the following voltage ranges and alternating current frequencies: 110-125 volt, 60 Hertz; 254-290 volt, 60 Hertz; or 210-236 volt, 50 Hertz. The particular voltage range and alternating current frequency, one of those cited herein, shall be specified (see 3.2.4 and 6.1). Additionally, ballast(s) shall be of the rapid start, high power factor type, having protective and automatic reset features of the thermostatic type, Class "P", meeting UL requirements. Ballast(s) shall be held firmly in contact with the body of fixture by at least 4 thread cutting (TC) sheet metal screws or by 4 non-turning studs, with nuts and lockwashers or equivalent used to secure the ballast to studs. Each luminaire shall be equipped with the following number and type of ballast(s) per lamp arrangement, to accomodate ANSI Standard C78.700 lamps:

Arrangement A - 1 two lamp ballast
Arrangement B - 1 two lamp ballast and 1 single lamp ballast
Arrangement C - 2 two lamp ballasts
Arrangement D - 3 two lamp ballasts
Arrangement E - 4 two lamp ballasts

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3.8 Plastic and glass material. Plastic material shall be 100 percent acrylic in accordance with ASTM D-702 or D-788. The rate of burning shall not exceed 3.2 inches per minute for a 1/2 inch width of plastic when tested in accordance with 4.5.1.1. Test results indicating compliance with ASTM D-702 or D-788 shall be required (see 4.5.1). Glass lenses for bottom closures and side panels (if used) shall have all edges "swiped" to remove sharpness.

3.8.1 Lens, diffuser and side panel marking. Each lens, diffuser (see 3.4.4.2), and side panel shall be permanently marked in an inconspicuous manner, with an identification which includes the following: manufacturer's name or trademark and coding or lens number. As an alternate to marking lens, diffuser, and side panel, marking may be placed elsewhere on fixture provided that such marking is as described above and placed in an inconspicuous area.

3.9 Finish on metal parts.

3.9.1 Reflector. Unless otherwise specified (see 6.1), the reflector surface finish shall be of synthetic enamel having a dry film thickness adequate to completely hide a blace and white substrate (see 4.5.3). Additionally, finish shall show no indication of chipping, cracking, flaking or any other sign of loss of adhesion when tested as specified in 4.5.3.

3.9.1.1 Reflectance. When determined in accordance with 4.5.3, the average reflectance of test samples after exposure shall not decrease by more than 4 percent of the initial average reflectance before exposure. Additionally, the average reflectance of test samples after exposure shall be at least 96 percent of average reflectance value given in independent laboratory photometric test data (see 4.6.6). Also when tested in accordance with 4.5.4, fixture average reflectance shall be at least 96 percent of the average reflectance value given in photometric test date (again, see 4.6.6).

3.9.2 Nonreflecting surface. Unless otherwise specified (see 6.1), the finish on all nonreflecting exterior surfaces shall be aluminum oxide on aluminum; white, gray, or aluminum paint or steel; nickel or chromium plating on copper alloy. Fastening devices shall be nickel, chromium, cadmium, or zinc plated. All painted surfaces shall be free from tears, star marks, blisters, black specks, hairlines, chipping, pinholes and other defects that may impair appearance or serviceability.

3.10 Salt spray test. When specified (see 6.1), one complete fixture, less the electrical components and bottom closure shall be subjected to the salt spray test specified in 4.5.2. Upon completion of test there shall be no evidence of corrosion.

3.11 Electromagnetic interference-free requirements. When specified (see 6.1), luminaire(s) shall be designed for suppression of electromagnetic interference such that requirements of MIL-STD-461 are complied with. Testing for electromagnetic interference shall be in accordance with 4.5.6.

3.12 Fixture rigidity. The assembled fixture, with required number of lamps (see 1.2) inserted therein, shall be subjected to the test specified in 4.5.5. Testing shall be accomplished with and without bottom closure installed. When tested in the manner outlined, neither fixture nor lamps shall become damaged or disassembled, bottom closure shall not drop open and all component parts of fixture shall remain in normal functioning positions.

3.13 Interchangeability of parts. The parts of each class, type, size, mounting, and lamp arrangement fixture of the same model number produced by the same manufacturer shall be interchangeable.

3.14 Fixture marking. Unless otherwise specified (see 6.1), each fixture shall be marked in accordance with provisions of UL-57. Marking shall be discernable when fixture is assembled but shall not be visible from the floor. Additionally, marking shall include FSN number, contract or order number, date (month and year) of manufacture and model number.

3.15 Photometric requirements.

3.15.1 Light distribution. When determined in accordance with 4.6.1, the percent of light distribution, expressed as a percentage of the fixture's total output in lumens, shall be as follows: 0-10 percent upward; 100-90 downward.

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3.15.2 Efficiency. Luminaires shall have the minimum efficiencies as indicated in table I when tested in accordance with 4.6.2. Efficiency as used herein is the ratio, in percent, of luminous flux leaving a fixture to the total luminous flux emitted by the lamp or lamps used therein. When calculating the efficiency of Class I and II fixtures, only the lumen output in the 0 to 90 degree zones shall be used; i.e., lumen output in the 90 to 180 degree zones shall not be included in efficiency computations.

TABLE I. EFFICIENCY

Class	Size	Lamp Arrangement	Efficiency (Minimum)
I	1	A	50
I	2	A	65
I	2	B	50
I	2	C	50
I	4	C	65
I	4	D	55
I	4	E	50
II	1	A	50
II	2	A	70
II	2	B	55
II	2	C	50
II	4	C	65
II	4	D	55
II	4	E	50
III	1	A	60
III	1	B	50
III	2	A	65
III	2	B	60
III	2	C	55
III	4	C	60
III	4	D	60
III	4	E	55
IV	1	A	50
IV	2	A	65
IV	2	B	50
IV	2	C	50
IV	4	C	65

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TABLE I. EFFICIENCY (Con't)

Class	Size	Lamp Arrangement	Efficiency (Minimum)
IV	4	D	50
IV	4	E	50
V	1	A	50
V	1	B	50
V	2	A	55
V	2	B	55
V	2	C	50
V	4	C	65
V	4	D	60
V	4	E	55

The efficiencies cited herein shall be as indicated regardless of Mounting or Type (see 1.2).

3.15.3 Visual comfort probability and luminance (class I, II, and IV luminaires only). When determined in accordance with the procedures specified in 4.6.4, fixture visual comfort probability (VCP) shall be 70 or more for crosswise and lengthwise orientations, VCP being based upon a room size of 60 feet x 60 feet and fixture mounting height of 10 feet above floor. Additionally, when determined in accordance with 4.6.3, the ratio of maximum to average luminaire luminance (photometric brightness) shall not exceed five to one at angles of 45°, 55°, 65°, 75°, and 85°, from nadir in the crosswise and lengthwise planes; and maximum luminance in the crosswise and lengthwise planes shall not exceed the following values:

Angle Above Nadir (degrees)	Maximum Footlamberts
45	2250
55	1605
65	1125
75	750
85	495

3.15.4 Spacing to mounting height ratio (class I, II and IV luminaires only). The spacing-to-mounting height (S/MH) ratio is the ratio of the distance between luminaire centers to the mounting height above the work plane, reference page 1-18 of IES Lighting Handbook, fifth edition 1972. When required, luminaire(s) shall be designed for a specified S/MH ratio(s) (see 6.1). The ratio(s), when specified, shall be determined in accordance with 4.6.5.

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3.16 Preproduction sample. Unless otherwise specified (see 6.1), before production is commenced, a sample fixture of each class, type, size, mounting and lamp arrangement specified in the Invitation for Bids shall be made available to the contracting officer or his authorized representative for approval in accordance with paragraphs 4.4, 4.5, 4.6, and 4.7. Test data, indicating that all required tests specified or described in paragraphs 4.5, 4.6, and 4.7 have been accomplished and indicating that all requirements of 3.3.1.2.7, 3.8, 3.9, 3.10, 3.11, 3.12, and 3.15 have been satisfied must accompany each preproduction sample at the time it is made available to the Government. Test data indicating compliance with 3.8, 3.9, 3.10, and 3.12 shall be no older than 12 months. Test data indicating compliance with 3.15 shall be no older that five years and must be obtained through an independent test laboratory acceptable to the Government. Other test data, required above, must be obtained through a laboratory acceptable to the Government. The requirements of this paragraph shall be applicable on a per contract basis.

3.17 Workmanship. Finished fixtures shall be clean and conform to the quality and grade of the product established by this specification. All burrs and rough and sharp edges shall be removed. Exposed welds or seams shall be ground smooth. All holes shall be accurately and cleanly punched or drilled. Fixtures shall be free from any defects that affect appearance or serviceability or any defects that may cause injury.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified (see 6.1), in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities for any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.

4.2.1 Preproduction sample inspection. The preproduction sample, shall be examined and tested as indicated in paragraph 3.16 above. Evidence of compliance shall be presented in the form of certified test reports.

4.2.2 Quality conformance inspection. Samples selected in accordance with 4.3.1 and 4.3.2 shall be examined or tested in accordance with 4.4, 4.5.2, 4.5.4 and 4.5.5.

4.2.3 Inspection of preparation for delivery. An inspection for preparation for delivery shall be performed in accordance with 4.8.

4.3 Sampling and acceptance levels for quality conformance inspection.

4.3.1 Sampling and acceptance levels for visual examination.

Sampling for visual examination (see 4.4) shall be performed in accordance with provisions set forth in MIL-STD-105. The Inspection Level shall be level II and except as noted below, the Acceptable Quality Level (AQL) shall be 2.5 percent for major defects, and 6.5 percent for minor defects, expressed in terms of percent defective.

4.3.2 Sampling and acceptance levels for tests. Sampling and quality acceptance levels for tests shall be as specified for each test in 4.5.2, 4.5.4 and 4.5.5.

4.3.3 Definition of lot. Lot as used herein shall be the collection of units of product from which samples are to be drawn and inspected to determine compliance with specification requirements. As regards fixtures, a lot shall consist of units of product of a single class, type (as applicable), size, mounting and lamp arrangement (see 1.2).

4.4 Visual examination. The completely assembled end item shall be examined for, but not limited to, the defects listed in table II. The inspection level and AQL shall be as specified in 4.3.1.

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4.4.1 Classification of defects.

TABLE II. Classification of defects

Defect	Requirement	Classification Major	Classification Minor
Parts not accessible	3.2.1	X	
Lamp spacing, not as specified	3.2.2	X	
Lampholders, not as specified	3.2.3	X	
Electrical requirements, not as specified	3.2.4	X	
Air supply, air return and heat exchange design not as specified	3.3.1.2.1, 3.3.1.2.2	X	
Construction not as specified	3.4	X	
Grounding device not as specified	3.4.1	X	
Wiring, not as specified	3.4.2	X	
Housing or chassis, not as specified	3.4.3	X	
Bottom closure, side panels not as specified	3.4.4	X	
Mounting not as specified	3.4.5	X	
Mounting hardware, not furnished	3.4.5.1 or 3.4.5.2		X
Dimensions, not as specified	3.4.6	X	
Material not as specified	3.5	X	
Parts incomplete	3.6	X	
Assembly not as specified	3.6	X	
Ballast(s), not as specified	3.7	X	
Plastic and glass material, not as specified	3.8	X	
Marking not as specified	3.8.1	see note	see note
Finish, not as specified	3.9.2	X	
Parts not interchangeable	3.13	X	
Marking, not as specified	3.14		X
Workmanship, not a specified	3.17	X	

Note: Lens, diffuser and panel (if used) identification marking (see 3.8.1) shall be compared with the identification information on photometric test report (see 4.6.5). If information on test report does not correspond with respective identification marking on lens, diffuser, and panel the entire fixture lot shall be rejected. Sampling shall be as specified in 4.3.1, no defects shall be allowed.

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4.5 Tests.

4.5.1 Plastic. Test results indicating compliance with ASTM D-702 or D-788 shall be provided by plastic manufacturer for each type of plastic used in the manufacture of fixtures covered herein.

4.5.1.1 Rate of burning. The rate or burning shall be as indicated in 3.8 when tested in accordance with Fed. Test Std. No. 406, Method 2021. This test shall be performed once for each class, grade, or type of plastic used in the manufacture of fixtures furnished under this specification.

4.5.2 Salt spray test. Fixture shall withstand the salt spray test as specified in Method 811.1 of Fed. Test Method No. 151 for a period of 48 hours (see 3.10). Only one sample per lot shall be tested. Failure shall be cause for rejection of the lot.

4.5.3 Exposure test. Subject at least five sample plates, each flat, 4 inches wide by 6 inches long and representative of the reflector finish and material, to testing in the following manner. Determine average reflectance of sample plates by measuring the reflectance of each sample with a reflectometer or instrument providing equivalent results, and averaging the cumulative reflectance measurements. Then expose sample plates for one hundred hours to the radiation of a glass enclosed carbon arc lamp, such as a Fade-O-Meter, operating on a current of 13 + .5 amperes and electrical potential, at the arc, of 140 volts. Each sample plate shall be placed ten inches from the arc and the lamp so ventilated that the temperature at the sample(s) does not exceed 105°F. After exposure, again determine the average reflectance as outlined above. Compare the initial average reflectance to average reflectance after exposure and ascertain compliance with the decrease of no more than 4 percent requirement of 3.9.1.1. Also compare average reflectance value of test samples after exposure with average reflectance value given in photometric test data (see 3.9.1.1 and 4.6.6). After comparing average reflectance, determine hiding of black and white substrate in accordance with Method 4121 of Fed. Test Method Standard No. 141 and examine for loss of adhesion (see 3.9.1). This test need only be conducted for preproduction sample fixture(s) when a preproduction sample(s) is required, or only for the first lot of fixtures of each class, type, size, mounting and lamp arrangement specified in the Invitation for Bids, when a preproduction sample(s) is not required. Decrease in average reflectance of more than 4 percent, signs of loss adhesion on any sample, or failure of more than 40 percent of samples to comply with the hiding of black and white substrate shall be construed as test failure and shall be cause for rejection.

4.5.4 Average reflectance. Average reflectance shall be determined by averaging the reflectance values taken at five randomly selected points on reflector surface. The average value so obtained shall be at least 96 percent of the value given in independent test lab data (see 3.9.1.1 and 4.6.6). Acceptable Quality Level (AQL) shall be 2.5, expressed in percent defective, in accordance with MIL-STD-105. Inspection Level shall be level II. Reflectance shall be measured by a reflectometer or instrument providing equivalent results.

4.5.5 Rigidity test. The fixture shall be lifted by application of a vertical force at one corner until such time as lift corner has been raised at least one foot and three of the four fixture corners are clear of horizontal support surface; the fourth corner (diagonally opposite lift corner) only remaining on the support surface. When in the correct test position the fixture shall be supported only at the lift corner and corner diagonally opposite. The test position shall be maintained for at least one minute whereupon compliance with requirements of 3.12 shall be determined. The above test shall be repeated until the lift force has been applied, in turn, at each of the four fixture corners. Four samples per lot shall be tested. Failure of any sample shall be cause for rejection of the lot.

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4.5.6 Electromagnetic interference. When specified in the Invitation for Bids (see 6.1), electromagnetic interference suppression tests shall be conducted in accordance with MIL-STD-461, and 462: Refer to MIL-STD-463, as required, for definitions and systems of units used in electromagnetic interference technology. The sample for this test shall be one complete fixture of each class, type, size, mounting and lamp arrangement specified in the Invitation for Bids. Testing shall be accomplished by test laboratory acceptable to the Government. This test need only be conducted for preproduction sample fixture(s) when a preproduction sample(s) is required, or only for the first lot of fixtures. Failure shall be cause for rejection of the lot. (see 3.11).

4.6 Light distribution, luminance, efficiency, visual comfort probability and spacing-to-mounting height ratio. The following tests shall be conducted on one fixture of each class, type, size, mounting and lamp arrangement specified in the Invitation for Bids. Additionally, all tests must be conducted by an independent test laboratory acceptable to the Government. (see 3.15).

4.6.1 Light distribution. In accordance with procedures outlined on pages 4-16, 4-17 and 4-18 of the IES Lighting Handbook, Fifth Edition (1972) under the general section titled "Photometry of Luminaires" and subsections titled "General Lighting Luminaires (Candlepower Distribution and Luminance (Photometric Brightness))"; and "Calculations of Luminaire Lumens From Candlepower Date", perform the following:

(a) Obtain fixture candlepower distribution data in at least 5 equally spaced vertical planes, to include the vertical planes that are parallel and perpindicular to the longitudinal axis of fixture. Candle power measurements, within each plane, shall be taken on both sides of fixture from nadir to 180 degrees above nadir, at intervals of 10 degrees or less, in the center of each interval (zone).

(b) From the above candlepower data, compute average luminous flux distribution in each zone from nadir to 180 degrees above nadir by averaging the zonal candlepower distributions of corresponding zones on both sides of fixture and multiplying by appropriate zonal constant.

(c) Compute the percentage of light distribution utilizing the luminous flux values calculated above and the following relations:

$$\text{Percent Upward Light Distribution} = \frac{\text{Lumen Output (90° to 180° zones)} \times 100}{\text{Lumen Output (0° to 180° zones)}}$$

$$\text{Percent Downward Light Distribution} = \frac{\text{Lumen Output (0° to 90° zones)} \times 100}{\text{Lumen Output (0° to 180° zones)}}$$

4.6.2 Efficiency. Fixture efficiency shall be determined in accordance with either of the procedures outlined on page 4-18 of the above mentioned IES Handbook under the subsection titled "General Lighting Luminaires (Total Lumen Output)". See 3.15.2.

4.6.3 Luminance (applicable to class I, II and IV Luminaires only. In accordance with procedures cited on pages 4-17, 4-18, and 4-25 of the above mentioned IES Handbook, under subsections titled "General Lighting Luminaires (Candlepower Distribution and Luminance (Photometric Brightness))" and "Luminance (Photometric Brightness) Measurements"; and procedures cited in Journal of IES, Volume I, No. 2 (IES Reprint LM-37) January, 1972 beginning on page 181, titled "Determination of Luminance of Luminaires", determine the maximum and average luminance for the angles and planes cited in 3.15.3.

4.6.4 Visual Comfort Probability (VCP) (applicable to class I, II and IV Luminaires only). Visual Comfort Probability (VCP) shall be determined in accordance with the procedures outlined in the Illuminating Engineering Society's "RQQ Report No. 2 (1972)-Outline of a Standard Procedure For Computing Visual Comfort Ratings For Interior Lighting" (IES Reprint LM-42). VCP calculations shall be based on a room size of 60 feet x 60 feet, fixture mounting height of 10 feet above floor level, and the standard conditions cited in above RQQ Report No. 2 (1972). See 3.15.3.

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4.6.5 Determination of spacing-to-mounting height ratio (applicable to Class I, II, and IV luminaires only). Spacing-to-mounting height ratio(s) shall be determined in accordance with procedures outlined in Fig. 9-14, on page 9-34, of the IES Lighting Handbook, Fifth Edition (1972). See 3.15.4.

4.6.6 Photometric tests and test data. Any changes made to the luminaire which would in any way affect any requirement in 3.15 shall require the performance of new photometric tests in accordance with 4.6. Data as furnished in compliance with 3.15 and 3.16 shall contain average fixture reflectance value, lens, diffuser, and side panel (if used) identification, as cited in 3.8.1, and shall be no older than five years. Date (month and year) of testing shall be indicated on all test reports. Average fixture reflectance shall be determined as specified in 4.5.4. All photometric tests shall be based on assumption that each lamp contained within the fixture produces 3100 lumens rated initial output. If lamps used for tests do not produce initial rated output, candlepower measurements and data shall be adjusted so that they equal measurements and data that would be produced at rated initial output. See 1.2 for lamp designations. All photometric tests outlined above (in 4.6) shall be accomplished under static non air-handling conditions.

4.7 Air supply, return, heat exchange and noise tests (applicable to class I luminaires only). Testing to determine compliance with the air supply/return, heat exchange and noise criteria requirements specified in 3.3.1.2.7 shall be in accordance with procedures outlined in Air Diffusion Council Test Codes TC-66R1 and 1062R3, as appropriate. Test data must indicate that air handling equipment functions or will function as required per 3.3.1.2, when used in conjunction with class I fixture(s) of the type, size, mounting and lamp arrangement specified in the Invitation for Bids. Any change to fixture or air handling equipment which would in any way affect any requirement in 3.3.1.2 shall require additional test data, indicating that performance requirements have not been compromised.

4.8 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table III. For examination of interior packaging, the sample unit shall be one shipping container prepared for delivery and selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

TABLE III. Classification of preparation for delivery defects

Examine	Defects
Marking (exterior and interior).	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Materials.	Any component missing or damaged.
Workmanship.	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of the container.

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5. PREPARATION.

5.1 Packaging. Packaging shall be level A, B or C, as specified (see 6.1).

5.1.1 Level A. Each fixture with accessories shall be packaged in a close-fitting box conforming to PPP-B-636, class weather-resistant variety, DW, Style FTC. Suitable blocking and bracing shall be utilized within the box to prevent movement or damage to the fixture or accessories. The box shall be closed and water-proofed sealed in accordance with the appendix to PPP-B-636.

5.1.2 Level B. Each fixture shall be packaged, as specified in 5.1.1, except that the box shall be in accordance with PPP-B-636, class-domestic and the water-proof sealing shall not be required. Closure of boxes shall be in accordance with method II of PPP-B-636.

5.1.3 Level C. Each fixture shall be packaged in accordance with the manufacturers standard practice, providing this insures protection for the fixture during shipment and safe delivery to its destination.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. Fixtures packaged, as specified in 5.1.1, shall be packed in quantities specified (see 6.1), in close-fitting boxes conforming to PPP-B-585, class 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-640, class 2. The box shall be closed and strapped in accordance with the applicable specification or appendix thereto. The gross weight shall not exceed 200 lbs.

5.2.2 Level B. Fixtures packaged, as specified in 5.1.2, shall be packed in quantities specified (see 6.1), in close-fitting boxes conforming to PPP-B-636, class-domestic variety DW, grade 275, style FTC. The box shall be closed in accordance with the appendix to PPP-B-636. The gross weight shall not exceed the weight limitations of the box specification.

5.2.3 Level C. Fixtures packaged, as specified in 5.1.3, shall be packed in containers that will assure carrier acceptance and safe arrival at destination in compliance with the Uniform Freight Classification Rules or the National Motor Freight Classification Rules.

5.3 Marking. Marking shall be in accordance with 5.3.1 or 5.3.2, as specified (see 6.1).

5.3.1 Military agencies. In addition to any special markings required by the contract or order, all interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Civil agencies. In addition to any special markings required by the contract or order, all interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

6. NOTES.

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, symbol, and date of this specification.
(b) Lamps, if required (see 1.1).
(c) Class, type, size, mounting, and lamp arrangement desired (see 1.2).
(d) Voltage and frequency requirements (see 3.2.4 and 3.7).
(e) Air handling function (in accordance with UL 57 Classifications A,B,C,D,E, or F for air handling fixtures) and Mode(s) of operation (applicable to class I fixtures only, see 3.3.1.2).
(f) Accessory slot covers, if not required (Class I only, see 3.3.1.2.3).
(g) If insulated supply air diffusers required and thickness of insulation (see 3.3.1.2.4) (Applicable to Class I luminaires only).
(h) Type ducting system(s) to be accommodated (see 3.3.1.2.4). (Applicable to class I luminaires only).

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(i) Air supply, air return, heat exchange and noise criteria requirements (see 3.3.1.2.7). (Applicable to class I luminaires only).
(j) If fixture manufacturer is to furnish air handling equipment (Class I only see note below para. 3.3.1.2.6 and paragraph 3.6).
(k) If design for mounting to combustible low-density sheet or tile ceilings is desired. (Applicable to class IV and V luminaires, see 3.3.2).
(l) If tempered glass is desired (see 3.4.4.1).
(m) Specific mounting requirements if desired (see 3.4.5.1 and 3.4.5.2).
(n) Specific tolerances for length and width dimensions, if desired (see 3.4.6).
(o) Luminaire height if other than indicated (see 3.4.6).
(p) Finish, if other than specified (see 3.9.1 and 3.9.2).
(q) Salt spray test, if desired (see 3.10).
(r) Electromagnetic interference-free requirements, if desired (see 3.11).
(s) Marking if other than specified (see 3.14).
(t) Spacing-to-mounting height ratio(s) if required (see 3.15.4).
(u) Preproduction sample, if not required (see 3.16).
(v) Responsibility for inspection, if other than indicated (see 4.1).
(w) Packaging and packing level required (see 5.1 and 5.2).
(x) Marking required (see 5.3).

6.2 Other fixtures. When required, refer to latest version of Federal Specification W-F-414 for coverage of pendant mounted office fixtures and latest version of specification W-F-1234 for coverage of industrial fixtures.

Military Coordinating Activity:

Army - ME

Preparing activity:

GSA - FSS

Civil Agencies Coordinating Activity:

COM - NBS
DC GOVT - DCG
DOT - FIS
HEW - NIH
INT - BPA
POSTAL - POS
VA - VOC

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